Plant and Soil **145**: 167–175, 1992

PLSO 9243

Transformations and plant uptake of urine-sulphate in urine-affected areas of pasture soil

P. H. WILLIAMS and R. J. HAYNES
MAF Technology, Canterbury Agriculture and Science Centre, P.O. Box 24, Lincoln, Canterbury, New Zealand

Received 7 June 1991. Revised December 1991

Key words: macropore flow, pasture soils, soil sulphur, sulphur transformations, sulphur fractionation

Abstract

The fate of sheep urine sulphate in the soil and its plant uptake was monitored using ^{35}S-labelled sulphate-S in undisturbed pasture microplots in two glasshouse experiments. The extent of macropore flow of simulated urine immediately following a sheep urination was also investigated at 5 pasture sites in the field.

Immediately following urination to pasture microplots in the glasshouse, the amounts of urine-derived ^{35}S recovered in the 0–2.5, 2.5–7.5, 7.5–15 and 15–30 cm soil layers were 38, 28, 18 and 9%, respectively. In the field study on 5 contrasting soils, a similar pattern was found with 55–70, 20–35 and 13–20% of simulated urine being recovered in the 0–5, 5–10 and 10–15 cm soil layers, respectively. There was insignificant loss below 15 cm. If urine had moved via simple displacement in these soils the wetting front would have reached only 2.0–2.5 cm in depth suggesting that significant downward movement of urine via macropore flow occurs after urination.

In a 15-day period following urine application to a pasture soil there was a rapid rate of incorporation of ^{35}S into organic forms, while between 15 and 64 days the rate of incorporation declined. After 7 days, 27% of added ^{35}S had been incorporated into organic forms with 19% being C-bonded S and 8% HI-reducible S. This rapid incorporation was attributed to the large and active microbial biomass present in the rhizosphere. Since urine application depressed pasture growth, due to 'urine burn', less than 10% of applied ^{35}S was absorbed by pasture plants over a 64-day period. A second experiment using microplots of contrasting soil types, confirmed that the majority of the ^{35}S incorporated into the organic form was present as C-bonded S. Results showed that of the ^{35}S remaining in the 0–2.5 cm layer 35 days after application, 20–40% was present as sulphate, 10–20% as HI-reducible S and 50–60% as C-bonded S. Plant uptake of S accounted for only 7–12% of applied ^{35}S over the 35-day period.

Introduction

The amount of S annually taken up by plants in intensively grazed pastures is often considerably larger than the rate of fertiliser S applied to maintain pasture production. The reason for this is that large quantities of S, as well as other nutrients, are cycled within the ecosystem through the actions of the grazing animals. Approximately 80–90% of the herbage S ingested by grazing animals is returned to the soil in the form of dung and urine. Commonly, 50–60% of ingested S is excreted in urine and 60–70% of this is present in the form of sulphate-S (Barrow, 1987; Bird, 1972). Large amounts of N, in the form of urea, and ionic K are also present in urine (Barrow, 1987).

Nutrients excreted in urine are applied to relatively small volumes of soil in quantities that often exceed the short-term requirements of the pasture plants growing in the urine patch. Also physical losses of nutrients can occur through downward macropore flow of urine immediately following application (Williams et al., 1990a) and other losses may occur later by leaching. The transformations, and pasture uptake of urine N (Thomas et al., 1988; Whitehead and Bristow, 1990) and K (Williams et al., 1990a,b,c) in the urine patch have been studied extensively. In contrast, the fate of urine S has received little attention, even though applications of fertiliser S are often required to maintain pasture production in New Zealand and elsewhere. In a field trial, Kennedy and Till (1981) found that over a 180-day period pasture plants were only able to recover 20% of sheep urine-S. Furthermore, some 40% of the applied S was unaccounted for and presumably leached. Rapid immobilisation of added sulphate into organic forms and its subsequent mineralisation are also known to be important pathways in the soil S cycle (Freney et al., 1975; McLaren et al., 1985). Indeed, in a laboratory study, using previously sieved soil, Boswell (1983) observed that 68% of urine S was incorporated into organic forms within 6 days.

This study investigated in field-like conditions the fate of ^{35}S-labelled urine-sulphate in the soil and its plant uptake in undisturbed pasture microplots in glasshouse experiments. The extent of macropore flow in a sheep urine patch was also estimated in a field study.

Materials and methods

Experiment 1

Undisturbed soil microplots (20 cm in diameter and 30 cm in depth) were collected from an established grass/clover pasture (mainly *Lolium perenne* L. and *Trifolium repens* L.) on Templeton Research Station near Christchurch, New Zealand. The soil was a Templeton silt loam (Typic Ustochrept, USDA) (Kear et al., 1967) and some relevant properties are presented in Table 1.

The microplots were collected by first exposing a soil monolith slightly larger than the final size of the sample, then gently forcing a rigid cylinder fitted with a cutting ring over the monolith. Molten vaseline was poured into the annular gap created by the cutting ring between the cylinder and the soil thereby preventing edge-flow of liquids down the outside of the soil in these experiments. The undisturbed microplots were detached from the soil and transported to a glasshouse. The bottom of each microplot was covered with nylon gauze, and the cores were weighed. The soil moisture content at the time of collection was at field capacity and throughout the experiment the soil moisture content was maintained at this level by daily applications of water. During the experiment mean soil temperatures (5 cm depth) in the microplots ranged from 14–21°C with a mean of 17°C. These temperatures are similar to mean soil temperatures

Table 1. Some properties of the soils used in the experiments

Soil	Depth (cm)	Total S ($mg\ kg^{-1}$)	C-bonded S ($mg\ kg^{-1}$)	Organic C (%)	pH (water)
Experiment 1					
Templeton	0–2.5	340	217	3.8	5.9
	2.5–7.5	305	176	2.4	5.7
	7.5–15	260	109	1.7	5.6
	15–30	222	186	0.84	5.8
Experiment 2					
Templeton	0–2.5	421	261	3.4	5.6
Temuka	0–2.5	1115	716	7.3	5.4
Waimakariri	0–2.5	496	340	3.7	5.9
Wakanui	0–2.5	413	289	4.4	5.8
Oxford	0–2.5	462	321	3.6	6.0

(5 cm) recorded in the field at a nearby meteorological station.

Urine (organic C content 2,000 mg L^{-1}, total S content 220 mg L^{-1}, sulphate-S content 190 mg L^{-1}) previously collected from sheep and then frozen, was thawed and mixed with carrier-free aqueous ^{35}S (9.3 MBq L^{-1}). Herbage on the soil microplots was trimmed to a height of 2 cm and 200 mL of urine were applied to 18 replicate microplots. In order to closely simulate sheep urinations the liquid was poured from a plastic bottle with a 1-cm diameter opening held at a height of 20 cm above the soil surface. This gave an approximate surface application rate of 6 L m^{-2}. A further 6 microplots were treated with 200 mL of distilled water (control treatment).

Three urine-treated microplots and a water-treated one were sampled at 1 hour (day 0), 1, 3, 7, 15 and 64 days following application. Herbage on the microplots was harvested, dried and ground. The soil was then removed from the cylinder, the vaseline scraped off, and the soil sectioned into 0–2.5, 2.5–7.5, 7.5–15, and 15–30 cm layers. The soil was then weighed, separated from any visible roots, sieved (<2 mm), thoroughly mixed and subsamples were air-dried and ground (<150 μm). Root material separated from the soil was washed, oven-dried and also ground.

Experiment 2

In this experiment 5 sites, which had been under permanent grass/clover pasture for more than 20 years, were sampled in the Canterbury region of New Zealand. The aim of this experiment was to determine whether the soil type affected the transformations of urine sulphate. Thus the sites were chosen to provide a range of contrasting soils, *viz*. Templeton silt loam, Temuka clay loam (Mollic Haplaquept), Waimakariri silt loam (Fulventic Ustochrept), Wakanui silt loam (Aquic Ustochrept) and Oxford silt loam (Aeric Fragiaquept). Some properties of these soils are shown in Table 1 and other details are given by Kear et al. (1967).

At each site a bromide recovery technique, developed to measure losses of urine by macropore flow following cattle urinations (Williams et al., 1990b), was used to measure the loss from sheep urinations. To simulate sheep urinations, 200 mL of KBr were poured onto the ground from a height of approximately 20 cm. After waiting twenty minutes for the solution to soak into the ground, 64 soil cores (15 cm in depth and 2 cm in diameter) were collected from an 8 × 8 grid pattern over each 'urine' patch area. Cores were sectioned into 3 layers (0–5, 5–10 and 10–15 cm) and cores from each layer were bulked together and analysed for Br (Abdalla and Lear, 1975). Five replicates of the experiment were carried out at each of the sites.

Four metal cylinders (15 cm in depth) were inserted into the soil at each site. Sheep urine (total S content 250 mg S L^{-1} and sulphate-S content of 170 mg S L^{-1}) labelled with ^{35}S (9.3 MBq L^{-1}) was applied to three of the cores at a rate of 150 mL per core. The remaining core received 150 mL of distilled water. Since a vaseline seal was not applied between the edges of the microplot and the metal cylinder, preferential flow of applied liquid down the edge of the microplot probably occurred. About 30 minutes after treatments were applied, the cores were dug carefully out of the ground and transported to a glasshouse. Soil moisture content was maintained at field capacity by daily water applications. After 35 days, the herbage was removed from the cores and the soil was then removed from the microplot cylinders. The 0–2.5 cm layer of soil was separated, visible roots removed, sieved (<2 mm) and a subsample was air-dried and ground. Washed herbage and root samples were dried, weighed and ground.

Analytical methods

Phosphate-extractable sulphate was extracted from the moist soil with 0.03 *M* $Na_2H_2PO_4$ (1:5 soil:solution ratio for 16 h) within 4 hours of soil sampling. The sulphate was measured after reduction to H_2S with the colorimetric methylene blue method (Tabatabai, 1982). Total S (by dry-ashing) and hydriodic acid (HI)-reducible S were measured on air-dried soil samples by procedures described by Tabatabai (1982). Carbon-bonded S was determined as the difference between total and HI-reducible S. HI-reducible organic S was determined as the difference between Hl-reducible S and phosphate-extractable S. Total S contents of urine and herbage samples were determined as outlined above after digestion with nitric and perchloric acids.

The activity of ^{35}S in the extracts used for ^{32}S analysis was determined using a liquid scintillation counter. One-mL samples from the extracts were mixed with 10 mL of scintillation cocktail (6 g 2, 5-diphenyloxazole dissolved in 670 mL of toluene and 330 mL of triton X) before counting. The activity in all the samples was corrected for radioactive decay to the start of each experiment.

Results

Experiment 1

Total recovery of applied ^{35}S at each sampling time varied from 88% ± 4 to 105% ± 4. Following application the proportion of ^{35}S recovered in sulphate form declined rapidly (Fig. 1) as S was incorporated into organic forms. After 7 days, 27% of ^{35}S was recovered in organic forms (19% as C-bonded and 8% as Hl-reducible S) and by 64 days this had risen to 37% (26% as C-bonded S and 11% as Hl-reducible S). The percentage of S absorbed by the roots and tops of pasture plants was small accounting for less than 10% of applied ^{35}S. This low uptake was attributable to a depression in pasture growth caused by 'urine burn' which is a reasonably frequent phenomenon on pastures. Pasture dry matter yields over the 64-day period were 1.0 and 1.5 g per microplot in urine and control treatments, respectively.

The recovery of ^{35}S as sulphate-S at different depths in the soil microplots at 0, 7 and 64 days is shown in Figure 2. The largest recovery of sulphate-^{35}S was in the 0–2.5 cm layer of soil and decreased with depth. Immediately after application (T = 0) 38, 28, 18 and 9% of ^{35}S were recovered in the 0–2.5, 2.5–7.5, 7.5–15 and 15–30 cm soil layers, respectively. Throughout the experiment a decrease in sulphate-^{35}S occurred mainly in the 0–2.5 cm soil layer and to a lesser extent at greater depths. The build-up of C-bonded ^{35}S (Fig. 3) was consequently most obvious in the 0–2.5 cm layer and least marked in the 15–30 cm layer. Data for Hl-reducible ^{35}S at various depths showed a pattern similar to that for C-bonded ^{35}S and are not presented.

Immediately following urine application there was an increase in sulphate-^{32}S levels in the soil (Fig. 4) that was largest in the 0–2.5 cm layer but also marked in the 2.5–7.5 cm layer. With time the sulphate-^{32}S content declined in both control and urine-treated soil due to a combination of plant S uptake and S immobilisation. For example, in the 0–7.5 cm soil layer the quantity of sulphate-^{32}S declined by 17.9 mg S per microplot in the control and 44.1 mg S per microplot in urine treatments between day 0 and 64. Corre-

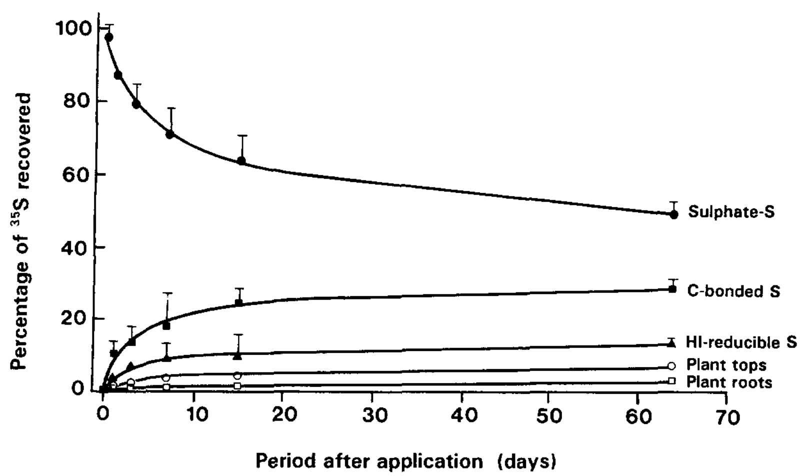

Fig. *1*. Recovery of ^{35}S labelled urine sulphate in the soil and herbage of pasture microplots (30 cm deep) during a 64-day period following application. Error *bars* represent SE (n = 3) and are not shown when smaller than symbols.

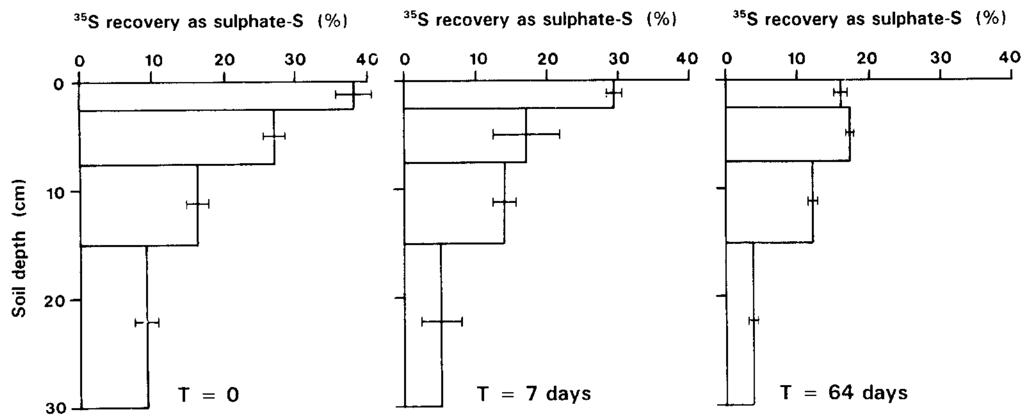

Fig. 2. Recovery of ^{35}S-labelled urine sulphate in sulphate-S form in the soil profile of pasture microplots at three times following application. Error *bars* represent ±SE (n = 3).

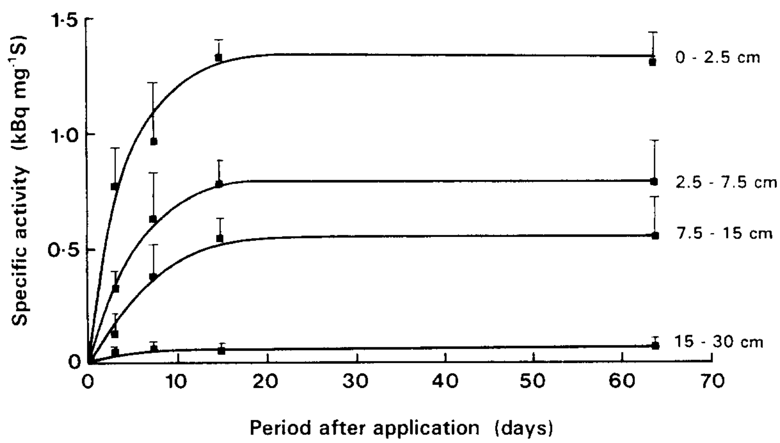

Fig. 3. Specific activity of ^{35}S in the C-bonded S fraction of soil from four layers of pasture microplots during a 64-day period following application. Error *bars* represent SE (n = 3).

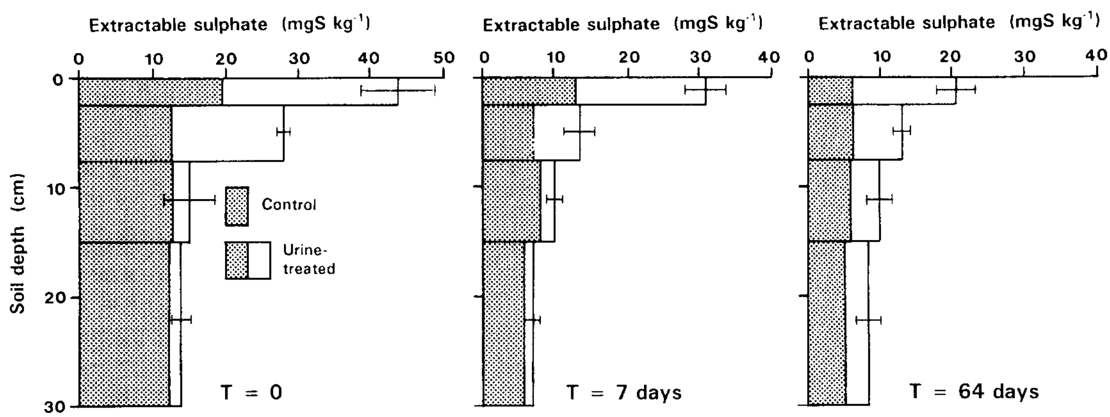

Fig. 4. Concentrations of sulphate-S in the soil profile from control and urine-treated pasture microplots at three times following application. Error *bars* represent ±SE (n = 3) for urine-treated cores.

sponding values for sulphate uptake by the pasture herbage for control and urine treatments were 6.1 and 3.4 mg S per microplot, respectively, suggesting that net immobilisation was larger than plant uptake. The specific activities of the sulphate-S pool in the soil at 0 and 64 days were similar (23 and 20, 15 and 18, 10 and 9, and 1 and 1 kBq mg S^{-1} at 0 and 64 days in the 0–2.5, 2.5–7.5, 7.5–15 and 15–30 cm depths respectively) which further suggests that there was negligible net mineralisation of native soil organic ^{32}S and so net immobilisation occurred over the experimental period.

No differences were found in ^{32}S content of the total S, HI-reducible S or C-bonded S fractions between urine-treated and control microplots (data not presented). Thus the high background levels of ^{32}S in these fractions prevented detection of any change caused by addition of urine S without the use of ^{35}S.

Experiment 2

Recovery of Br in the soil profile from simulated urine patches at 5 separate sites is shown in Figure 5. The pattern of recovery was broadly similar at all five sites with about 55–70% recovered in the 0–5 cm layer, 25–35% in the 5–10 cm layer and 13–20% in the 10–15 cm layer. Movement of Br below 15 cm must have been insignificant since calculated recovery in the 0–15 cm layer was 103%, 114%, 111%, 112% and 96% for the Templeton, Temuka, Waimakariri, Wakanui and Oxford soils, respectively. All sites had similar moisture contents (0.25–0.3 g g^{-1}).

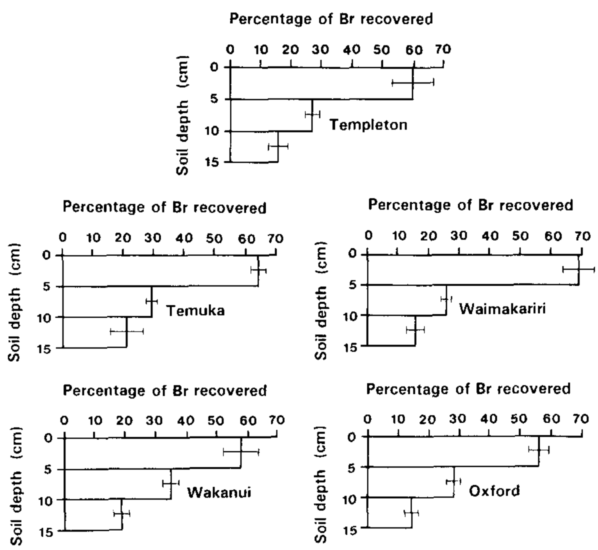

Fig. 5. Field recovery of Br in the soil profile from sheep urinations simulated using potassium bromide solution at five pasture sites. Error *bars* represent ±SE (n = 5).

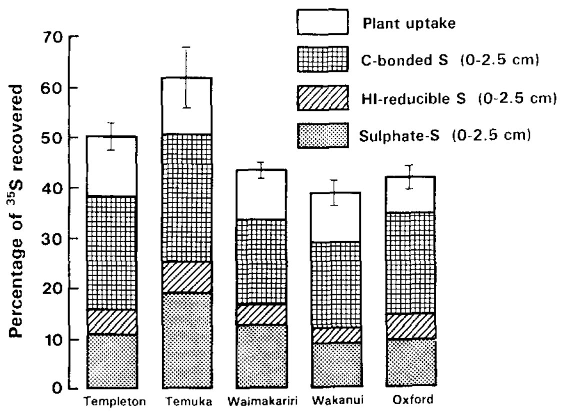

Fig. 6. Recovery of ^{35}S-labelled urine sulphate in the surface soil (0–2.5 cm) and herbage of pasture microplots 35 days after application. Error *bars* represent ±SE (n = 3) for total recovery of ^{35}S in soil and herbage.

The recovery of applied ^{35}S in the 0–2.5 cm soil layer and in the plant herbage from microplots from the five sites is shown in Figure 6. Between 33 and 50% of applied S remained in the 0–2.5 cm layer of soil. Of this S, 20–40% was present as sulphate-S, 10–20% as HI-reducible S and 50–60% was C-bonded S. Plant uptake accounted for 7–12% of applied ^{35}S. Herbage dry matter yields were not greatly affected by urine application, except for an increase in the Waimakariri soil. Herbage dry matter yields for control and urine-treated cores were 3.90 and 3.40, 4.23 and 4.61, 2.60 and 4.21, 2.37 and 2.37, 1.92 and 1.60 g per microplot for the Templeton, Temuka, Waimakariri, Wakanui and Oxford soils, respectively.

Discussion

Following each sheep urination significant preferential flow of urine down soil macropores was observed. This was evidenced by the measurement of ^{35}S in all soil layers down to 30 cm immediately after urine application in experiment 1. Similarly, in the Br recovery experiments 15–20% of Br was recovered in the 10–15 cm soil layer, whereas if urine had moved down the soil profile by simple displacement, the wetting front would have reached a depth of just 2.0–2.5 cm. Visual observations using methylene blue dye as a urine tracer confirmed that urine moved down the surface opening macropores that had been created by soil cracks, plant roots and earthworms.

Despite preferential flow, most of the urine in the field experiments was retained in the 0–15 cm layer with insignificant movement below 15 cm. In experiment one, 9% of urine ^{35}S was calculated to have moved below 15 cm. However, the flow of urine in an undisturbed soil core (used in experiment 1) may differ from that in the field where sideways movement of liquids is not restricted by the presence of the cylinder walls. These results differ from those of Williams et al. (1990b) who found that for cattle urinations 15–25% of urine often flows down macropores to below a 15-cm soil depth. The magnitude of macropore flow is likely to be less for sheep than for cattle urinations since the surface application rate of a sheep urination is half that of cattle. In view of the considerable variability in macropore flow that is commonly observed from site to site due to variability in soil physical properties (Williams et al., 1990b), results for Br recovered with depth were surprisingly consistent between sites. It seems that loss of urine below 15 cm, via macropore flow following sheep urinations, is not a major mechanism of nutrient loss from pastures.

Throughout the experiment a large proportion of the added sulphate remained in the soil in organic form. This may not be the case in the field where sulphate is susceptible to leaching. Plant uptake in the field may also be higher, as under field conditions there is often a marked positive pasture growth response in the urine-patch area (During and McNaught, 1961) which may result in rapid uptake of accumulated sulphate-S. In the present glasshouse study pasture growth was insignificant due to either urine burn or to short duration of the experiment.

In experiment 1, incorporation of S into organic fractions was very rapid, with 27% being incorporated within 7 days. A similarly rapid rate of incorporation of ^{35}S into organic fractions has been observed in laboratory incubation studies where rewetted, previously air-dried and sieved soils have been used (e.g. Ghani et al., 1988; McLaren et al., 1985). In such studies, the rapid ^{35}S incorporation has been attributed to a flush of microbial activity following rewetting. However, the present study used undisturbed, intact soil microplots with moisture contents maintained at field capacity. Since a ready supply of carbonaceous material exists in the pasture rhizosphere of these soils, the microbial biomass and microbial activity would have already been high (Jenkinson and Oades, 1979; Sarathchandra et al., 1984). For example, microbial biomass C at the experimental sites (measured by chloroform-fumigation and direct extraction) ranged from 850 to 1000 $\mu g\,C\,g^{-1}$. In addition, urine application may stimulate soil microbial activity since it contains a small amount of organic material. Thus these soil microplots were expected to have high microbial activity which could have resulted in a rapid net immobilisation of added ^{35}S. Similarly, net immobilisation of urea-N in the urine patch can also be rapid and Keeney and MacGregor (1978) observed that 13% of applied ^{15}N was incorporated into organic forms within 7 days. Since immobilisation and mineralisation occur concurrently, it is likely that during the experimental period some ^{35}S would have been immobilised and subsequently mineralised. That the rate of accumulation of ^{35}S into organic matter declined sharply in the period between 15 and 64 days indicated that in that period mineralisation almost balanced immobilisation.

Both experiments showed that applied ^{35}S was incorporated preferentially into the C-bonded fraction to a much greater extent than into the HI-reducible fraction. This is not surprising because C-bonded S was the major form of organic S present in the study soils (Table 1) and C-bonded S has been shown to be the major form of organic S that accumulates in soils during pasture development (McLachlan and De Marco, 1975). Even so, Maynard et al., (1985) observed, in incubation studies, that when solution sulphate levels were high, as in the case in urine patches, the majority of ^{35}S incorporation was into the HI-reducible S fraction. Preferential incorporation of ^{35}S-labelled sulphate into the

C-bonded S fraction has been observed when there is a burst of microbial activity, such as when a soil is rewetted and/or a readily available C source is added to the soil (Ghani et al., 1988; Maynard et al., 1985; McLaren et al., 1985). This has been attributed to a flush of microbial growth causing a shortage of S and the microbial biomass formed has a low Hl-reducible and high C-bonded S content (Maynard et al., 1985). Such a phenomenon apparently occurs because under high ambient sulphate levels soil fungi tend to accumulate ester sulphates as storage compounds, whilst under S stress they have a lower ester sulphate content and a higher proportion of their S is present in C-bonded form (Sagger et al., 1981).

The above explanation is not, however, applicable to the present study where in spite of large amounts of sulphate remaining in the urine patch throughout the duration of the experiments, S was preferentially accumulated as C-bonded S. As previously explained, there is likely to be a large active microbial biomass present in the soil microplots. Therefore it seems possible that under conditions of high microbial activity and a rapid turnover of microbial S (e.g. a pasture urine patch, a rewetted soil or a soil to which an available C source has been added) then S is preferentially incorporated into C-bonded forms (regardless of the ambient sulphate levels). On the other hand, where microbial activity and turnover is less rapid there may be a greater tendency for accumulation of S into ester sulphate forms.

Regardless of the reason, it seems that in the urine patch, urine sulphate accumulates as organic S principally in C-bonded forms. The initial incorporation of S into C-bonded forms does not necessarily imply that this is its final destination. Ghani et al. (1988) found that ^{35}S recently incorporated into C-bonded forms is particularly susceptible to remineralisation to sulphate and possibly Hl-reducible S. Similarly, when ^{35}S-labelled methionine is added to soils, the C-bonded S is rapidly converted to sulphate and Hl-reducible forms (Fitzgerald and Andrew, 1984; Fitzgerald et al., 1984). Thus, much of the S initially incorporated into the C-bonded fraction may be subsequently converted to sulphate and/or Hl-reducible S. Further studies will be required to measure remineralisation of the ^{35}S incorporated into organic forms.

Although soil organic S accumulates during pasture development, an equilibrium state is reached under a given pasture management regime after 20 to 50 years where S mineralisation and immobilisation are balanced (Jackman, 1964; Quin and Rickard, 1981). However, this does not preclude net immobilisation occurring in some areas of the pasture (e.g. newly formed urine patches) and net mineralisation occurring in other areas (e.g. areas not recently affected by urine).

Conclusions

The rapid incorporation of about 30% of added sulphate into an organic form in the urine patch suggests that about 30% of the deposited urine sulphate is cycling through organic pools in the soil, whilst the balance remains in inorganic form. That significant quantities of urine-derived sulphate were found to remain in the urine patch as sulphate suggests that the efficiency of S cycling will depend largely on whether conditions following urination favour rapid pasture growth, and hence S uptake, or leaching losses. Field studies under differing conditions will reveal the relative importance of these opposing processes to S cycling in the urine patch.

Immobilisation of urine sulphate-S in the urine patch was mainly into C-bonded S form and consequently S mineralisation must also occur predominantly from this fraction. Research to identify the nature of the dynamic pool of C-bonded S will greatly improve our understanding of S cycling under pasture.

Acknowledgements

We thank the Agricultural and Marketing Research and Development Trust (AGMARDT) for partial funding of the research and Mrs A M Lynch, Mr P L Carey, Miss R J Rubie and Miss L M Goodman for skilled technical assistance.

References

Abdalla N A and Lear B 1975 Determination of inorganic bromide in soils and plant tissues with bromide selective -ion electrode. Commun. Soil Sci. Plant Anal. 6, 489–494.

Barrow N J 1987 Return of nutrients by animals. *In* Managed Grasslands. Ed. R W Snaydon. pp 181–186. Elsevier, Amsterdam.

Bird P R 1972 Sulphur metabolism and excretion studies in ruminants. IX. Sulphur, nitrogen, and energy utilisation by sheep fed a sulphur-deficient and a sulphate-supplemented roughage-based diet. Aust. J. Biol. Sci. 25, 1073–1085.

Boswell C C 1983 An assessment of phosphorus and sulphur interactions on aspects of the sulphur cycle in pasture. Ph.D. thesis, University of New England, Australia.

During C and McNaught K J 1961 Effects of cow urine on growth of pasture and uptake of nutrients. N. Z. J. Agric. Res. 4, 591–605.

Fitzgerald J W and Andrew T L 1984 Mineralization of methionine sulphur in soils and forest floor layers. Soil Biol. Biochem. 16, 565–570.

Fitzgerald J W, Andrew T L and Swank W T 1984 Availability of carbon-bonded sulfur for mineralisation in forest soils. Can. J. Forest. Res. 14, 839–843.

Freney J R, Melville G E and Williams C H 1975 Soil organic matter fractions as sources of plant-available sulphur. Soil Biol. Biochem. 7, 217–221.

Ghani A, McLaren R G and Swift R S 1988 The incorporation and remineralisation of ^{35}S in soil organic fractions. *In* Towards the More Efficient Use of Soil and Fertiliser Sulphur. Fertiliser and Lime Research Centre Occasional Report No 2. Eds. R E White and L D Currie. pp. 32–46. Massey University, Palmerston North, N.Z.

Jackman R J 1964 Accumulation of organic matter in some New Zealand soils under permanent pasture. I. Patterns of change of organic carbon, nitrogen, sulphur and phosphorus. N. Z. J. Agric. Res. 7, 445–471.

Jenkinson D S and Oades, J M 1979 A method for measuring adenosine triphosphate in soil. Soil Biol. Biochem. 11, 193–199.

Kear B S, Gibbs H S and Miller R B 1967 Soils of the downs and plains of Canterbury and North Otago, New Zealand. Soil Bureau Bulletin No. 14. N. Z. Soil Bureau, Wellington.

Keeney D R and MacGregor A N 1978 Short-term cycling of ^{15}N-urea in a ryegrass-white clover pasture. N. Z. J. Agric. Res. 21, 443–448.

Kennedy A P and Till A R 1981 The distribution in soil and plant of ^{35}S from sheep excreta. Aust. J. Agric. Res. 32, 339–351.

Maynard D G, Stewart J W B and Bettany J R 1985 The effects of plants on soil sulphur transformations. Soil Biol. Biochem. 17, 127–134.

McLachlan K D and De Marco D G 1975 Changes in soil sulphur fractions with fertiliser additions and cropping treatments. Aust. J. Soil Res. 13, 169–176.

McLaren R G, Keer J I and Swift R S 1985 Sulphur transformations in soils using sulphur-35 labelling. Soil Biol. Biochem. 17, 73–79.

Quin B F and Rickard D S 1981 Pasture production and changes in soil fertility on a long-term irrigated superphosphate trial at Winchmore, New Zealand. Proc 14th Int. Grassl. Congr. 323–326.

Saggar S, Bettany J R and Stewart J W B 1981 Measurements of microbial sulfur in soil. Soil Biol. Biochem. 13, 493–498.

Sarathchandra S U, Perrott K W and Upsdell M P 1984 Microbiological and biochemical characteristics of a range of New Zealand soils under established pasture. Soil Biol. Biochem. 16, 177–183.

Tabatabai M A 1982 Sulfur. *In* Methods of Soil Analysis. Eds. A L Page, R H Miller and D R Keeney. pp 501–538. American Society of Agronomy, Madison, WI.

Thomas R J, Logan K A B, Ironside A D and Bolton G R 1988 Transformations and fate of sheep urine-N applied to an upland U.K. pasture at different times during the growing season. Plant and Soil 107, 173–181.

Whitehead D C and Bristow A W 1990 Transformations of nitrogen following the application of ^{15}N-labelled cattle urine to an established grass sward. J. Appl. Ecol. 27, 667–678.

Williams P H. Hedley M J and Gregg P E H 1990a The effect of preferential flow of dairy cow urine and simulated rainfall on movement of potassium through undisturbed topsoil cores. Aust. J. Soil Res. 28, 857–868.

Williams P H, Gregg P E H and Hedley M J 1990b Use of potassium bromide solutions to simulate dairy cow urine flow and retention in pasture soils. N. Z. J. Agric. Res. 33, 489–495.

Williams P H, Gregg P E H and Hedley M J 1990c Fate of potassium in dairy cow urine applied to intact soil cores. N. Z. J. Agric. Res. 33, 151–158.